

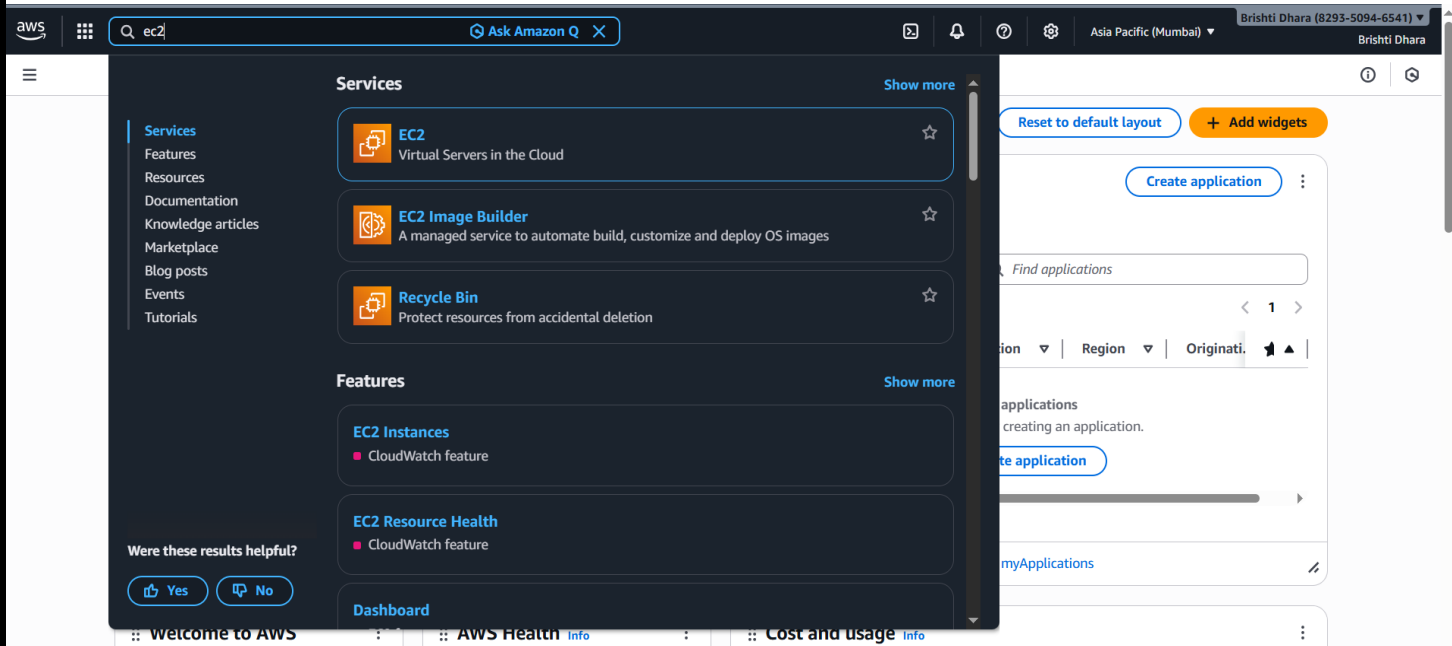
Name: BRISHTI DHARA Roll. No: 90 ExamRoll: 11600223043

AssignmentNo:7

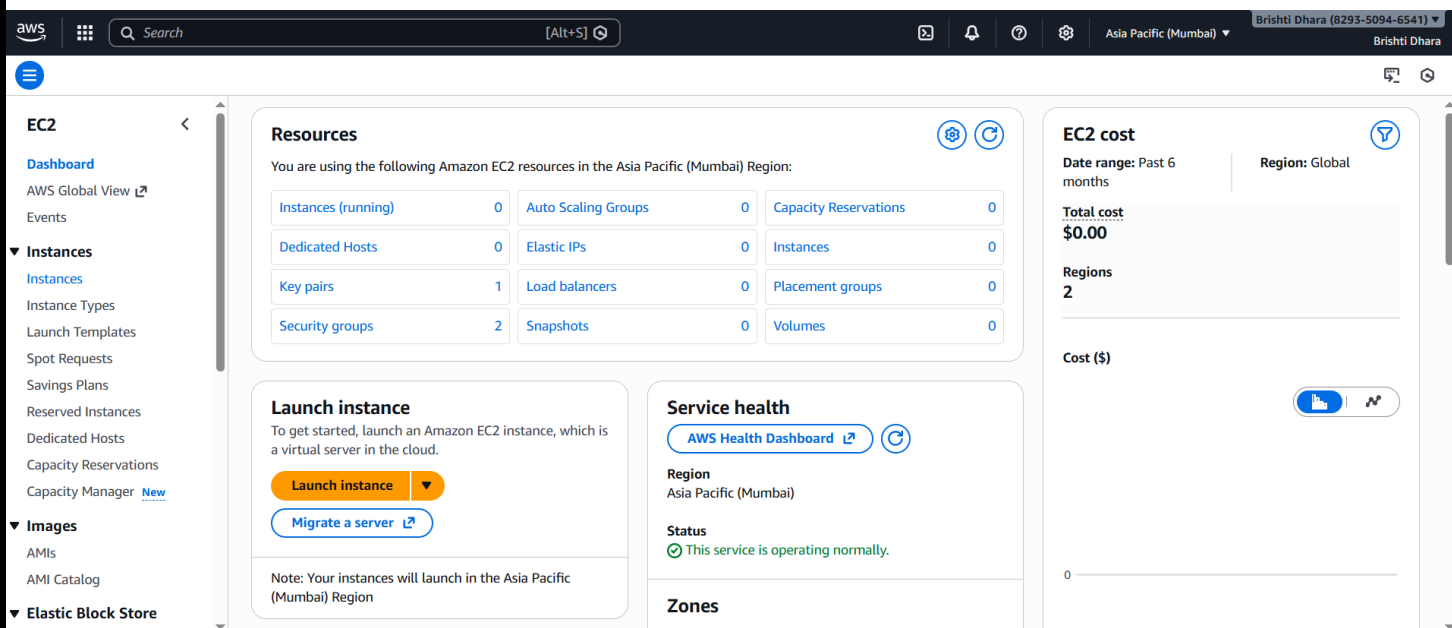
Problem Statement: Hosting a website on EC2.

Solution:

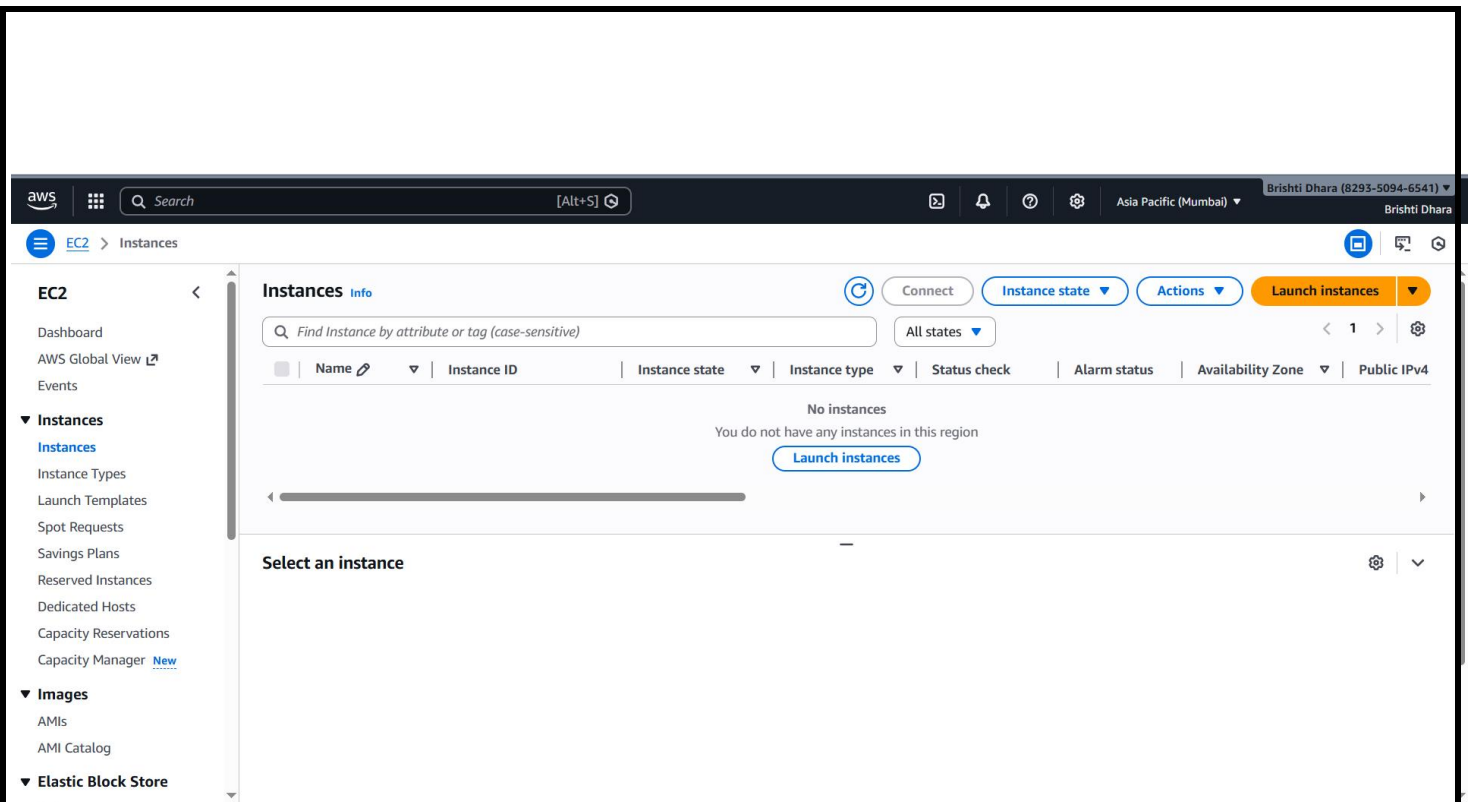
Step 1: Search EC2 in the search bar and click on it.



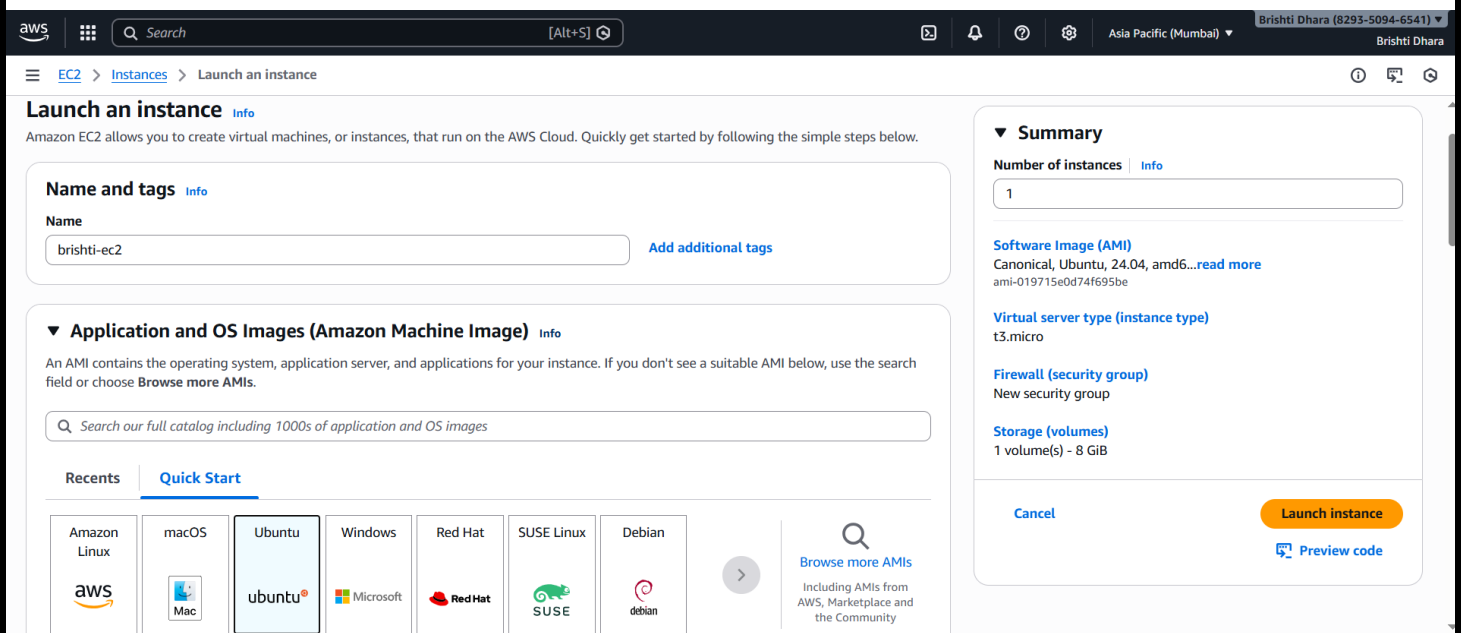
Step 2: On the page click **Instances**.



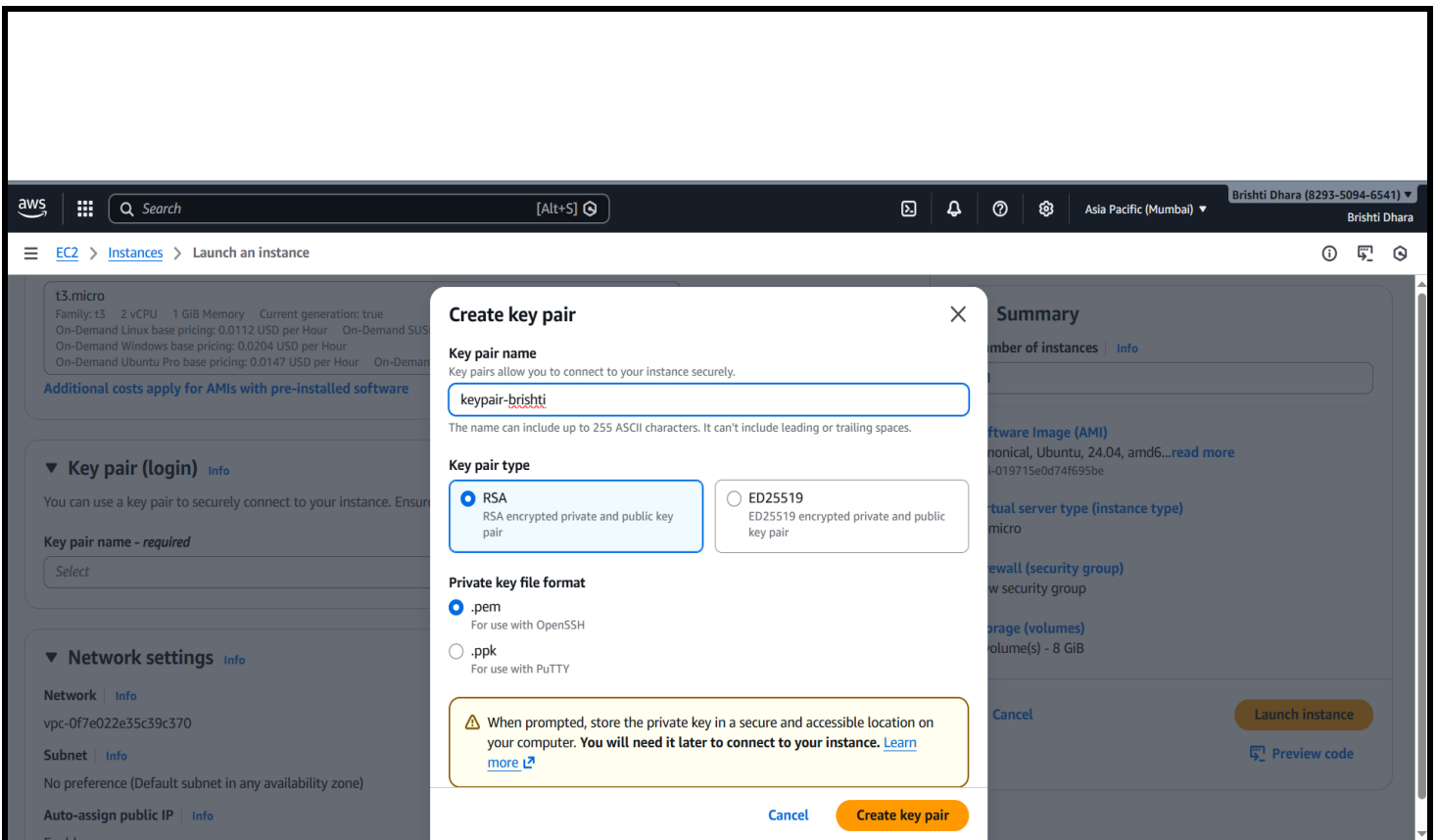
Step3: In the Instances page, click on **Launch instance** button.



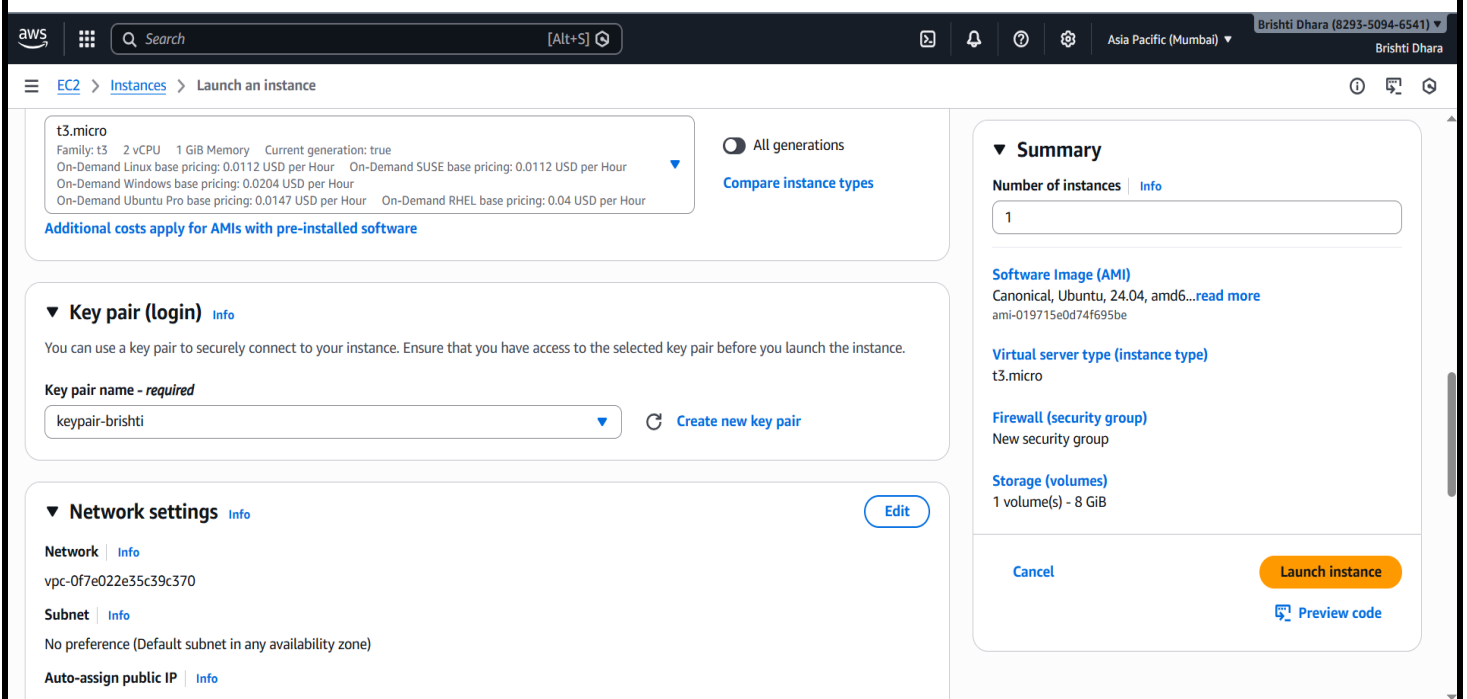
Step 4: Enter a name for the instance in this case **brishti_ec2**, select **Ubuntu** and then click on **Create new key pair**.



Step 5: Enter the name of Keypair, Select RSA and click on **Create key pair** button.



Step 6: Select the key from the drop down, also the key will be downloaded on the desktop.



Step7: Click on **Create security group** and check the boxes of **Allow HTTPS traffic from the internet** and **Allow HTTP traffic from the internet** and click on **Launch instance** button.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Brishti Dhara (8293-5094-6541)

Brishti Dhara

EC2 > Instances > Launch an instance

1

2

▼ Network settings

Info

Edit

Network

Info

vpc-0f7e022e35c39c370

Subnet

Info

No preference (Default subnet in any availability zone)

Auto-assign public IP

Info

Enable

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

We'll create a new security group called 'launch-wizard-2' with the following rules:

✓ Allow SSH traffic from

Helps you connect to your instance

Anywhere

0.0.0.0/0

✓ Allow HTTPS traffic from the internet

To set up an endpoint, for example when creating a web server

✓ Allow HTTP traffic from the internet

To set up an endpoint, for example when creating a web server

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-019715e0d74f695be

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

aws

Search

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Asia Pacific (Mumbai)

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Brishti Dhara

EC2 > Instances > Launch an instance

1

2

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

×

▼ Configure storage

Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🔄 Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

► Advanced details

Info

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-019715e0d74f695be

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Step 8: The Instance will be successfully created, return to the **Instance** page.

Success
Successfully initiated launch of instance (i-0b601340da4f5a799)

► Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
[Create EBS snapshot policy](#)

Step9:The Instance will be seen running, click on the newly created instance.

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
brishti-ec2	i-0b601340da4f5a799	Running	t3.micro	Initializing	View alarms	ap-south-1a	ec2-13-127-

i-0b601340da4f5a799 (brishti-ec2)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

Instance ID i-0b601340da4f5a799	Public IPv4 address 13.127.5.97 open address	Private IPv4 addresses 172.31.37.209
IPv6 address -	Instance state Running	Public DNS ec2-13-127-5-97.ap-south-1.compute.amazonaws.com

Step 10: Copy the **Public IPv4 address** and open the bitwise application on the computer.

The screenshot displays the AWS Management Console interface for the instance **i-0b601340da4f5a799 (brishti-ec2)**. The instance is in the **Running** state. The **Public IPv4 address** is **13.127.5.97**. The **Private IP address** is **172.31.37.209**. The **Instance type** is **t3.micro**. The **Auto-assigned IP address** is **13.127.5.97 [Public IP]**. The **Subnet ID** is **subnet-036bbf9173876782e**. The **Instance ARN** is **arn:aws:ec2:ap-south-1:829350946541:instance/i-0b601340da4f5a799**.

Step 11: Pasted the copied IPv4 address to the **Host** field, Enter **Ubuntu** in the Username field and select the Initial method to **publickey** and click on **Client key manager**.

The screenshot shows the Bitwise SSH Client 9.57 application window. The **Default profile** is selected. The **Host** field is set to **13.127.5.97**, the **Username** is **ubuntu**, and the **Initial method** is **publickey**. The **Client key** field is empty. The **Client key manager** button is highlighted. The **Log in** button is also visible.

Step 12: Click on **import** and select the downloaded key from the desktop.

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows navigation options like Dashboard, AWS Global View, Events, and Instances. The main content area shows the 'Instance summary' for instance ID i-0b601340da4f5a799. Overlaid on this are two windows from the Bitvise SSH Client 9.57. The top window, 'Client Key Manager', shows a table of SSH user authentication keys with columns for Location, Algorithm, Size, Password, SHA-256 Fingerprint, MD5 Fingerprint, Bubble Babble, and Comment. It includes buttons for 'Generate New', 'Modify', 'Remove', 'Import', 'Export', 'Change Passphrase', and 'More'. The bottom window, 'Select Client Key Import File', shows a file explorer view of the 'This PC > Documents' directory. It lists several folders and files, with 'keypair-brishti.pem' selected. The 'File name' field at the bottom of this window contains 'keypair-brishti.pem' and the file type is set to 'Private Key Files (*.bkey; *.ppk; *)'. The background console shows instance details such as IP address (13.127.5.97), Hostname type (IP name: ip-172-31-72.31.37.209), Answer private resource (IPv4 (A)), Auto-assigned IP address (13.127.5.97 [Public IP]), IAM role, and IMDSv2 (Required).

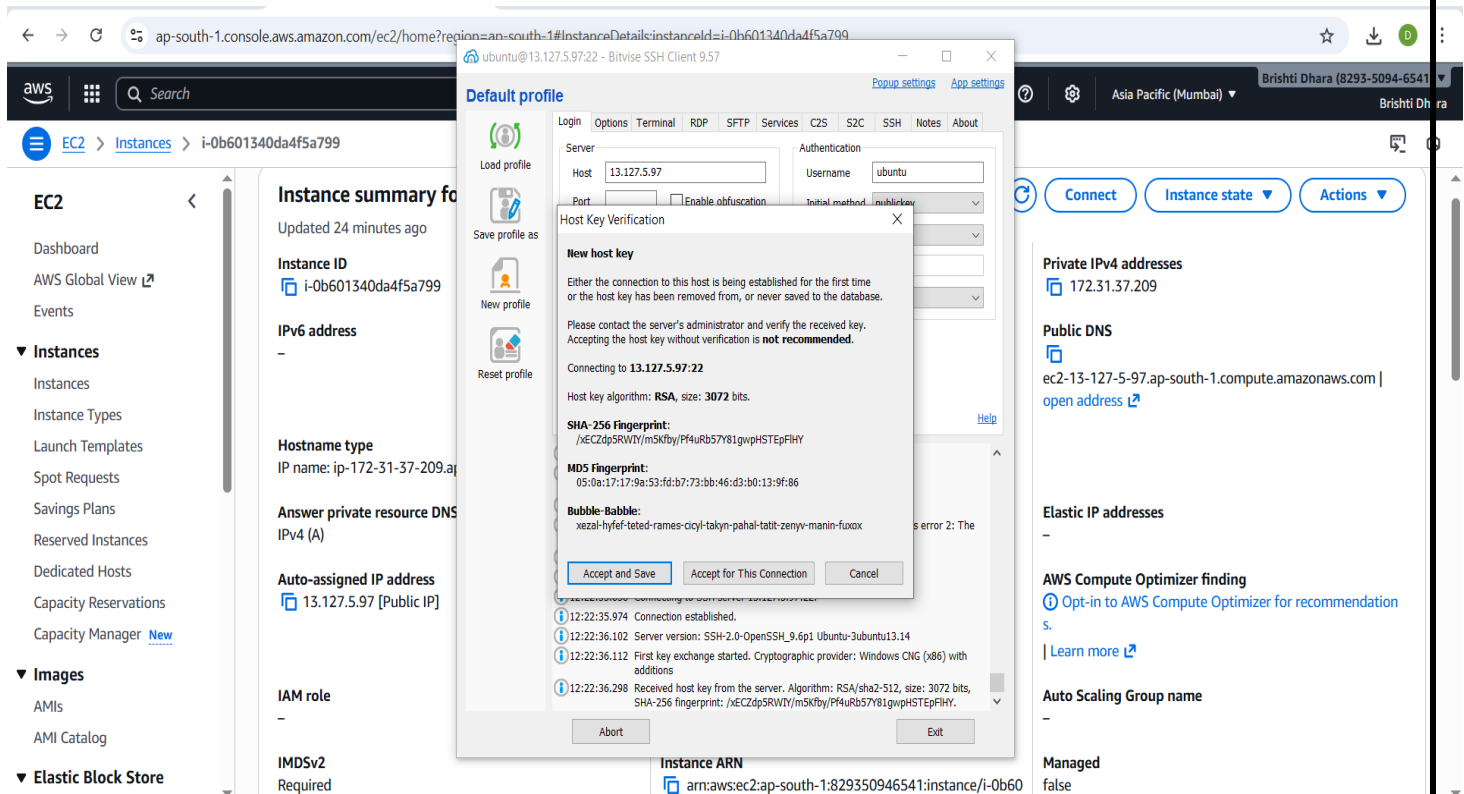
Step13:Client will be selected **Global1**, click on the login button.

The screenshot displays the AWS Management Console interface for an EC2 instance, with the Bitvise SSH Client 9.57 window overlaid. The AWS console shows the instance details for `i-0b601340da4f5a799` in the `ap-south-1` region. The instance is in the `Running` state. The Bitvise SSH Client window is configured with the following settings:

- Default profile:** Login, Options, Terminal, RDP, SFTP, Services, C2S, S2C, SSH, Notes, About.
- Server:** Host: `13.127.5.97`, Port: `22`, Enable obfuscation: ☐.
- Authentication:** Username: `ubuntu`, Initial method: `publickey`, Client key: `Global 1`, Passphrase: `Default`.
- Kerberos:** SPN: `SPN`, GSS/Kerberos key exchange: ☐, Request delegation: ☐, gssapi-keyex authentication: ☒.

The Bitvise SSH Client window also shows a log of system messages, including the current date (2026-02-14) and the client version (9.57). The AWS console shows the instance summary, including the instance ID, IPv6 address, hostname type, answer private resource DNS, auto-assigned IP address, IAM role, and IMDSv2 status.

Step 14: It will be logged in with the instance, click on the new terminal console button from the side panel.



Step15: Update the system using the command **sudo apt-get update && sudo apt-get upgrade**.

```
Last login: Sat Feb 14 06:54:12 2026 from 42.105.194.221
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-37-209:~$ sudo apt-get update && sudo apt-get upgrade
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1742 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [325 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.5 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1529 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [313 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.9 kB]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2594 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [595 kB]
```

Step 16: Install nginx into the linux system using the command **sudo apt-get install nginx**.

```
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for plymouth-theme-ubuntu-text (24.004.60-1ubuntu7.1) ...
update-initramfs: deferring update (trigger activated)
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for debianutils (5.17build1) ...
Processing triggers for initramfs-tools (0.142ubuntu25.8) ...
update-initramfs: Generating /boot/initrd.img-6.14.0-1018-aws
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...
systemctl restart acpid.service chrony.service cron.service irqbalance.service packagekit.service polkit.service rsyslog.service ssh.service udisks2.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #3: sshd[1068,1179]
ubuntu @ user manager service: systemd[1073]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-209:~$ sudo apt-get install nginx
```

Step17: Navigate to the **var/www/** folder and give **html** full permission using **chmod**.

```
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart networkd-dispatcher.service
systemctl restart serial-getty@ttyS0.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

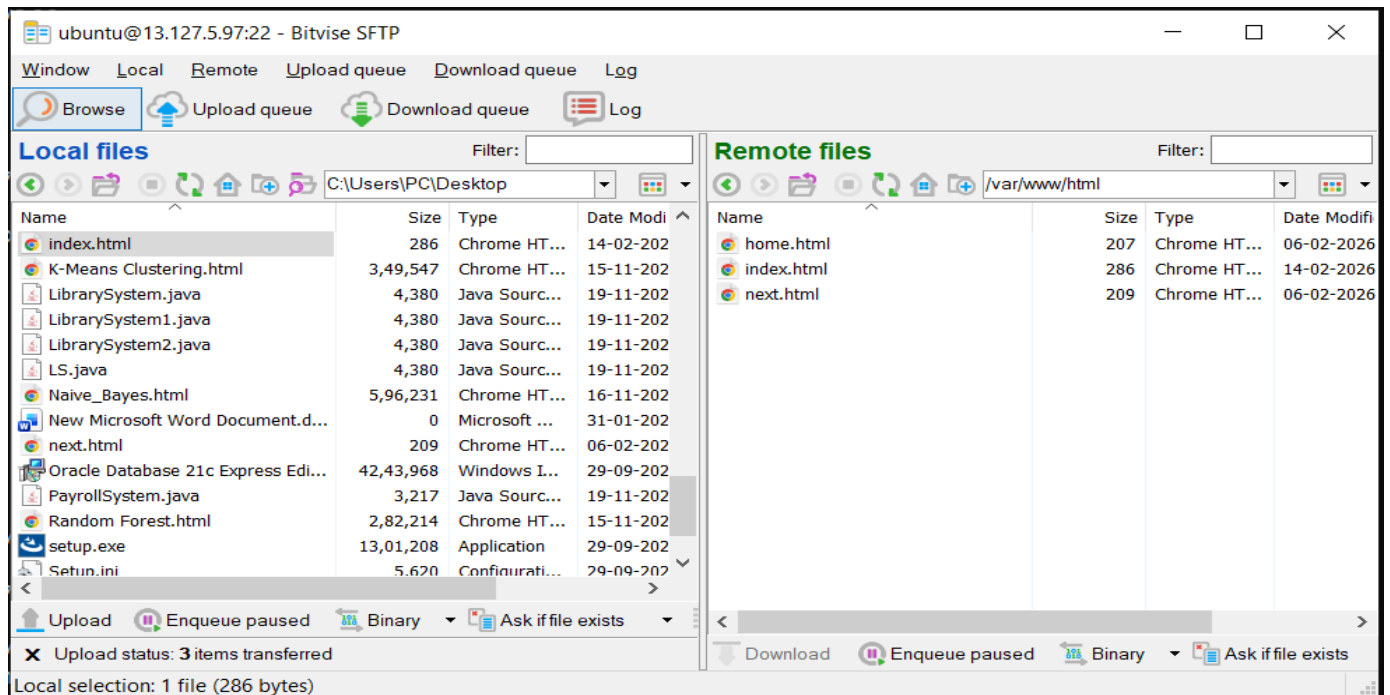
No containers need to be restarted.

User sessions running outdated binaries:
ubuntu @ session #3: sshd[1068,1179]
ubuntu @ user manager service: systemd[1073]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-37-209:~$ cd /
ubuntu@ip-172-31-37-209:/$ ls
bin          dev          lib          lost+found  opt         run          snap        tmp
bin.usr-is-merged etc        lib.usr-is-merged media       proc        sbin        srv         usr
boot         home        lib64       mnt        root       sbin.usr-is-merged sys         var

ubuntu@ip-172-31-37-209:/$ sudo chmod 777 /var/www/html
ubuntu@ip-172-31-37-209:/$ cd /var/www/html
ubuntu@ip-172-31-37-209:/var/www/html$ ls
home.html  next.html
ubuntu@ip-172-31-37-209:/var/www/html$ cd /var/www/html
ubuntu@ip-172-31-37-209:/var/www/html$ ls
home.html  index.html  next.html
ubuntu@ip-172-31-37-209:/var/www/html$
```

Step 18: In the bitwise click on New SFTP window, on the window, on the left side go to the directly where the static website is present and on the right window go to the directly `/var/www/html` then drag and drop the three files from the **Local files** to the **Remote Files**.



Step19: We can access the website from a new tab via the IPv4 address copied earlier.



Step-20: Go instance state and from the dropdown list select Stop instance.

The screenshot shows the AWS Management Console interface for the 'Instances' page. The left sidebar contains navigation links for EC2, Images, and Elastic Block Store. The main content area displays a table of instances. The instance 'brishti-ec2' (ID: i-0b601340da4f5a799) is in the 'Running' state. The 'Instance state' dropdown menu is open, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. The 'Stop instance' option is highlighted.

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status
brishti-ec2	i-0b601340da4f5a799	Running	t3.micro	3/3 checks passed

i-0b601340da4f5a799 (brishti-ec2)

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0b601340da4f5a799	13.127.5.97 open address	172.31.37.209
IPv6 address	Instance state	Public DNS
-	Running	ec2-13-127-5-97.ap-south-1.compute.amazonaws.com

Step 21: In the popup, click on the Stop button. Then instances Successfully initiated Stopping.

The screenshot shows the AWS Management Console interface for the 'Instances' page. A green notification banner at the top states 'Successfully initiated stopping of i-0b601340da4f5a799'. The instance 'brishti-ec2' (ID: i-0b601340da4f5a799) is now in the 'Stopping' state. The 'Instance state' dropdown menu is open, showing the 'Stop instance' option.

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
brishti-ec2	i-0b601340da4f5a799	Stopping	t3.micro	3/3 checks passed	View alarms	ap-south-1a	ec2-13-127-5-97.ap-south-1.compute.amazonaws.com

i-0b601340da4f5a799 (brishti-ec2)

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-0b601340da4f5a799	13.127.5.97 open address	172.31.37.209
IPv6 address	Instance state	Public DNS
-	Stopping	ec2-13-127-5-97.ap-south-1.compute.amazonaws.com

Step 22: We again select the same Instance and click on **Instance state** from the drop down we select **Terminate (delete) instance** option. Then the Instance will be successfully terminated.

The screenshot displays the AWS Management Console interface. At the top, a green notification banner states: "Successfully initiated termination (deletion) of i-0b601340da4f5a799". Below this, the "Instances" page is shown with a table of instances. The instance "brishti-ec2" (ID: i-0b601340da4f5a799) is highlighted, and its state is "Stopping". The instance type is "t3.micro" and it is in the "ap-south-1a" availability zone. The "Instance state" dropdown menu is open, showing the "Terminate (delete) instance" option selected. The "Details" tab is active for the selected instance, showing the instance ID, public IPv4 address (13.127.5.97), private IPv4 addresses (172.31.37.209), and public DNS (ec2-13-127-5-97.ap-south-1.compute.amazonaws.com).

EC2 > Instances

Successfully initiated termination (deletion) of i-0b601340da4f5a799

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
brishti-ec2	i-0b601340da4f5a799	Stopping	t3.micro	3/3 checks passed	View alarms	ap-south-1a	ec2-13-127-

i-0b601340da4f5a799 (brishti-ec2)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0b601340da4f5a799	Public IPv4 address 13.127.5.97 open address	Private IPv4 addresses 172.31.37.209
IPv6 address -	Instance state Stopping	Public DNS ec2-13-127-5-97.ap-south-1.compute.amazonaws.com